

JENNIFER F. WALDERN

DATA SCIENCE & AI ARCHITECT • AI AND PRODUCT EVALUATION & MEASUREMENT

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SUMMARY

Data scientist and AI architect with **15 years** across Microsoft and SAP, specializing in **AI and product evaluation, measurement, and experimentation**. **7+ years** in Windows data science on product experiences reaching **1B+ Windows 11 devices**, owning executive-facing metrics, forecasts, and roadmap inputs, including which features drive engagement and retention. Builds LLM evaluation harnesses, custom MCP servers, and multi-agent workflows that turn ambiguous problems into trusted metrics and AI systems teams adopt. Pursuing an MSc in Epistemology, Ethics and Mind at the University of Edinburgh.

SELECTED IMPACT

- Originated the **account-profile badging** feature space, tied to a projected **600M+ revenue/business opportunity**; badging is now live on the account avatar across **Windows, Edge, and Microsoft 365**.
- Built the File Explorer Search Session dataset and **Search Success Rate (v1-v3)** metric, supporting **15 feature rollouts** to 1B+ devices.
- Caught a user-habit regression that standard scorecards missed and **stopped a broad-population rollout** in Insider preview.
- Originated **"Windows Sells Windows"** and aligned cross-org stakeholders to launch it as an initiative **still in market today**.

CORE TECHNICAL SKILLS

AI Evaluation & Measurement: LLM evaluation harnesses, LLM-as-judge meta-evaluation, reliability gating, pre-registration and power analysis, human labeling and inter-rater agreement, A/B testing, difference-in-differences, metric design, scorecards, AI Value Index metrics.

AI Architecture & LLM Systems: LLM-assisted workflows, applied generative AI, custom agents, multi-agent teams, custom MCP server development, skills/hooks, prompt orchestration, retrieval workflows, Microsoft Copilot CLI and Copilot Chat, internal Microsoft agency plugins, Athena NLP query workflows, Claude, CLINE.

Machine Learning & MLOps: Azure Databricks, PySpark/Spark pipelines, Python ML workflows, feature engineering, model evaluation, classification, forecasting, anomaly detection, collaborative filtering, segmentation, churn analysis, model monitoring, lineage, governance, production readiness.

Data Engineering, BI & Platforms: Advanced SQL, Python, ScopeScript, R, Azure Synapse, Azure Data Studio, Azure ML Studio, Power BI, Jupyter, ETL/ELT pipelines, data modeling, analytics engineering, API integration, SAP HANA, SAP BusinessObjects BI Platform, SAP Data Services, CRM/ERP data, legacy-system migration, Jenkins, Git, Docker.

Product Analytics & Executive Decision Support: Microsoft Account MAD, DAD, and MAU; Click to Do Weekly Engagement Rate and onboarding/adoption metrics; File Explorer Search Success Rate; experimentation strategy; forecasting; roadmap prioritization; SLT/CVP/VP-level executive communication.

APPLIED AI, EVALUATION & ARCHITECTURE

- Designing a **pre-registered meta-evaluation of LLM-as-judge reliability** for grounded generation, with a contamination check, blind pairwise human labeling with intra-rater repeats, and an explicit power analysis; analysis plan published on GitHub.
- Developed **custom evaluation harnesses** from scratch to assess output correctness, consistency, quality thresholds, and workflow reliability across AI-assisted systems.
- Designed and implemented **LLM-assisted workflow pipelines** for structured decision support, including prompt orchestration, evaluation scoring, reliability gates, reusable review loops, and decision-state tracking.
- Built **agent-style architectures** integrating structured tools, retrieval workflows, custom MCP servers, skills, hooks, and multi-agent collaboration patterns.
- Evaluated when to use **traditional ML, automation, generative AI, or human-in-the-loop workflows** based on reliability needs, business impact, data quality, and adoption risk.

PROFESSIONAL EXPERIENCE

Senior Data Scientist – Microsoft

Windows Data Science | 2022 – Present

- Envisioned and advanced a new **account-profile gamification feature space**, including avatar badging, validating feature-impact hypotheses through experimentation and influencing **10+ initiatives** adopted across Windows 11 surfaces; tied to a projected **600M+ revenue/business opportunity**, with the badging paradigm now live across Windows, Edge, and Microsoft 365.
- Originated **"Windows Sells Windows"**, an enhancement to the Windows Trade-in program that brings trade-in opportunities directly to the user's device; aligned cross-org stakeholders to launch it as an initiative still in market today.
- Applied collaborative filtering, time-series forecasting, anomaly detection, user segmentation, classification, churn analysis, and onboarding-flow analysis to identify **growth opportunities** and improve product decisions.

- Owned forecast inputs and informed annual roadmaps from **FY21 through FY26** across Microsoft Accounts and later Click to Do, translating metric movement, adoption patterns, and product hypotheses into investment decisions.

File Explorer Search & File Agent (semantic, federated, lexical, indexed/unindexed) | Aug 2025 – Present

- Built the **File Explorer Search Session dataset**, a custom multi-stage data pipeline, and designed **File Explorer Search Success Rate (v1-v3)** on top of it: a multi-component metric measuring search success, click-through quality, click-back behavior, result-position weighting, and task completion.
- Caught a **user-habit regression** in the Downloads folder that standard scorecards missed while the change was still in Insider preview; **reversed an engineering push to ship**, stopping a broad-population rollout that would have degraded established user behavior and carried PR risk.
- Support **15 feature rollouts** to the broad Windows 11 population (1B+ devices) on that dataset, including A/B test design and metric setup, scorecard analysis, and deep-dive investigation.
- Analyzed the search relevancy ranking algorithm and ran user-behavior and **difference-in-differences analysis** across devices and sessions to evaluate a default sort-order change.

Modernize with AI — Click to Do (Copilot+ PC) | Jan 2025 – Aug 2025

- Created **AI Value Index metrics** and related frameworks to evaluate AI experiences across usage, engagement, success signals, and business value, supporting roadmap prioritization for emerging AI features.
- Supported analytics, onboarding/adoption measurement, and **Weekly Engagement Rate** evaluation for Click to Do, an AI-powered feature on Copilot+ PCs; identified priority funnel drop-offs and adoption opportunities.

Microsoft Accounts (MSA) & M365 | 2022 – Jan 2025

- Drove **YoY growth** for the Microsoft Account Key Result portfolio, including MAD, DAD, and MAU measurement; delivered recurring leadership updates, annual forecasts, target-setting analysis, anomaly detection, root-cause analysis, and roadmap recommendations.
- Redesigned Windows account measurement from **device-based to user-based attribution** by creating and validating the data pipeline, improving metric accuracy, sensitivity, and decision usefulness across partner organizations.
- Presented Microsoft Accounts data in Windows Business Review forums for **5+ years** to org-wide SLT audiences, including CVP- and VP-level leaders.
- Spearheaded metric definition alignment and logic harmonization with partner organizations, improving consistency, lineage, and decision quality across Windows measurement surfaces.
- Initiated and coordinated **additional IC resources** for the FY24 roadmap focused on personalization and the user-based data product, using data insights and feature gains to justify expanded investment.

Founder & Technical Lead – HitsHome

AI Decision Intelligence Platform – Real Estate & Location Analytics | 2024 – Present

- Architecting an **AI-powered decision intelligence platform** integrating structured data pipelines, scoring models, LLM-assisted workflows, and agent-enabled decision support for real estate and location analytics.
- Built **data ingestion and feature engineering pipelines** combining Census, housing, and economic data into standardized, decision-ready features across counties, places/cities, ZCTAs/ZIP codes, and states.
- Developing personalized market analysis, property matching, structured risk assessment, confidence scoring, and **explainable recommendation workflows** using agentic systems, schemas, custom tools, and evaluation patterns.

Data and Applied Scientist II – Microsoft

Windows Operating System, Core Windows Experiences | Dec 2018 – Jan 2022

- Functional manager and project lead for a **team of 3** through the full data science lifecycle: hypotheses, user research, prototyping, testing, implementation, experimentation, and retail feature launch.
- Provided coaching and feedback, completed performance review cycles, coordinated with university programs, and recommended full-time offers for all three interns.
- Managed cross-org data projects to **increase engagement and revenue** through the Microsoft Account Attach Rate metric and sub-metrics, machine learning clustering, exploratory insights, and multi-source data joins.
- Vetted and led experimentation shipment processes, finalized **green-light launch recommendations** with leadership, and used statistical analysis for scorecard creation and hypothesis testing.
- Built analysis and reporting workflows using Python, ScopeScript, Azure ML, Azure Data Studio, SQL, Power BI, and supervised/unsupervised methods.

Research Assistant – Microsoft Research

Ability Lab, Neurodivergence and Multimodal AI Design | Jan 2019 – Oct 2020

- Generated hypotheses, shaped the research vision, conducted experiments and interviews, coded and analyzed results, and wrote and edited academic text for Ability Lab collaborations.
- Focused on cognitive information processing and communication models in autistic populations, contributing to a **new framework and model** with Microsoft Research collaborators.
- Co-authored **two published ACM papers** in HCI and accessibility, bridging qualitative research and responsible AI.

Data Scientist; QA Developer; Product Owner – SAP

Enterprise Analytics | Apr 2015 – Oct 2018

- Applied CRM and ERP expertise to predictive modeling using SAP Predictive Analytics, Python, R, and Anaconda; built data models, ETL workflows, and reports in SAP HANA.
- Architected the migration strategy for SAP BusinessObjects Platform, including an audit of **200 data sources and 40K reports**; managed legacy-system dependencies and reduced migration risk.
- Implemented automated versioning for the BusinessObjects Cloud Library Extension and built a regression-identification framework with Jenkins and Docker.
- Implemented an Agile model for the Build Infrastructure team and **decommissioned 77 servers** (about 80% of internal DevOps application servers), including backup strategy, execution, and communication.
- Led creation of a full-stack analytics solution for SAP Product Idea Management, moving customer feature-request data from BrightIdea via PHP, Python, Selenium, and SAP Data Services into SAP HANA for **240 product managers across 50+ products**.
- Served as intern manager and hiring/onboarding lead for new full-time team members.

Global Project Lead; BI Tools Engineer; Database Developer – SAP

Development Operations | Sep 2011 – Feb 2015

- Led upgrades, testing cycles, performance work, and customer support for SAP BusinessObjects Enterprise 4.1 on a global internal instance serving **2,000+ users**.
- Addressed system warnings proactively to reduce customer impact, using Zabbix and Windows PerfMon.
- Led post-upgrade automated testing across data extraction, modeling, and reporting workflows using Java and Selenium.
- Led three co-op hiring rounds with four competitive hires, and led an SBOP global project team of six.
- Supported the DevOps unit through troubleshooting, administration, data extraction, security, monitoring, and data modeling across SBOP, IBM ClearQuest, JIRA, and Perforce CVS; built an audit database with SAP BusinessObjects IDT to improve event monitoring.

LEADERSHIP & COMMUNITY

- **6+ years** as team lead, intern manager, mentor, and project lead; conducted interviews, managed hiring and onboarding, supported performance reviews, and led teams of 2–6 across Microsoft and SAP.
- **8+ years** of product owner and analyst experience, including roadmap definition, stakeholder requirements, metric alignment, compliance considerations, risk communication, and data-driven product execution.
- **Co-Founder, CEO, and President – Girls in Tech Vancouver**: led strategy and operations, managed 20+ volunteers, served as board president, grew membership to **400+**, secured **30+ sponsors**, and ran 20+ events supporting the retention of women in technology.
- **Founder – SAP-UBC GIRLsmarts4tech**: established an SAP and UBC partnership teaching computer science to Grade 7 girls; grew the volunteer base **from 5 to 150 in one year** and supported seven program runs, including three international iterations; 400+ leadership volunteer hours.
- **3,000+ hours** of volunteering across Allen Institute consciousness research, the Windows Inclusivity Culture Team, Microsoft Research Ability Team, Bringing Women Together Network, UBC Sauder mentoring, design-thinking coaching, and youth education.

PUBLICATIONS & RESEARCH

- **“Managing Stress: The Needs of Autistic Adults in Video Calling”** – Proceedings of the ACM on Human-Computer Interaction, Volume 3, Issue CSCW, November 2019.
- **“Lessons Learned in Designing AI for Autistic Adults”** – ASSETS 2020: Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility, October 2020.
- **LLM Judge Trustworthiness: a pre-registered meta-evaluation** – analysis plan on GitHub (JennyFrancesWaldern/judge-meta-eval).
- **Coherence analysis of rodent MRI data** – Allen Institute consciousness and psilocybin research collaboration.

EDUCATION

- 2026 – Present · **University of Edinburgh**, School of Philosophy – **MSc in Epistemology, Ethics and Mind** (part-time, online).
- 2022 · **University of Washington**, Computational Neuroscience Center – Computational Neuroscience Research Topics; top of class.
- 2018 · **BrainStation**, Canada – Data Science and Modeling.
- 2010 – 2013 · **University of British Columbia**, Canada – **B.Sc., Major in Computer Science** with Co-op. Course highlights: Artificial Intelligence, Calculus, Linear Algebra, Databases, Cognitive Science, and Statistics.
- 2008 – 2010 · **University of Victoria**, Canada – B.Sc. studies in Computers, Mathematics, and Economics; transferred.